

Objective

Design a modular and scalable architecture for the Smart Lender application by separating the frontend, backend, machine learning model, and testing components. This improves code organization, maintainability, and scalability.

Three-Tier Architecture

1. Presentation Layer (User Interface)

The presentation layer is the frontend of the application where users interact with the system. It is developed using HTML, CSS, and Bootstrap.

Responsibilities:

- Display the loan application form.
- Accept applicant details.
- Submit data to the backend.
- Display the prediction result.

Files:

```
templates/  
  index.html  
  result.html
```

```
static/  
  style.css
```

2. Backend API Layer

The backend is developed using Flask.

Responsibilities:

- Receive user input.
- Validate the data.
- Preprocess the input.

- Load the trained XGBoost model.
- Predict loan approval.
- Return the result to the frontend.

File:

app.py

3. AI/ML Service Layer

This layer contains the trained machine learning model.

The following algorithms are trained:

- Decision Tree
- Random Forest
- K-Nearest Neighbors
- XGBoost

The XGBoost model is selected because it provides the best prediction accuracy.

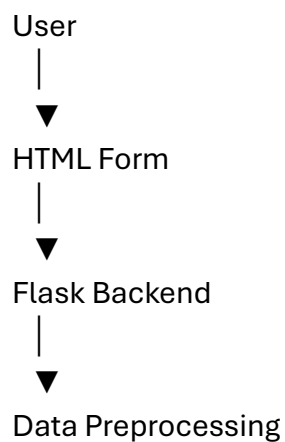
Files:

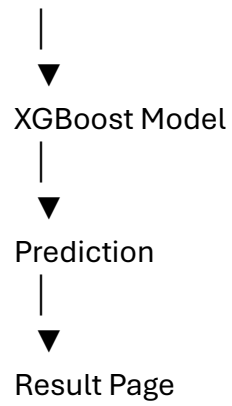
model/

model_training.py

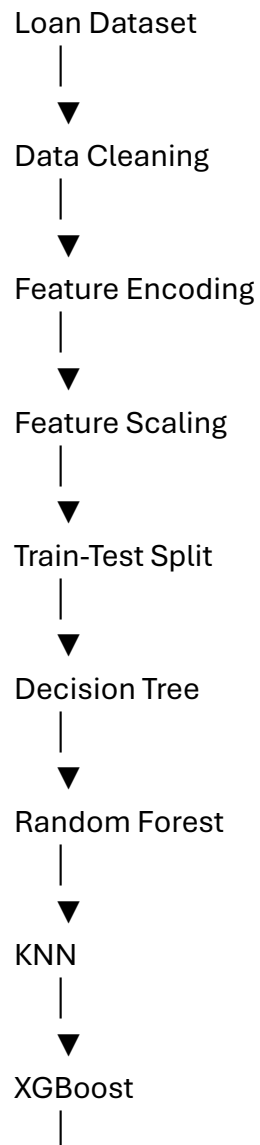
xgboost_model.pkl

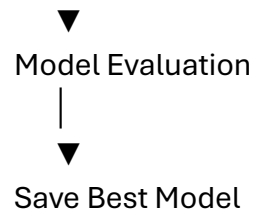
Data Flow



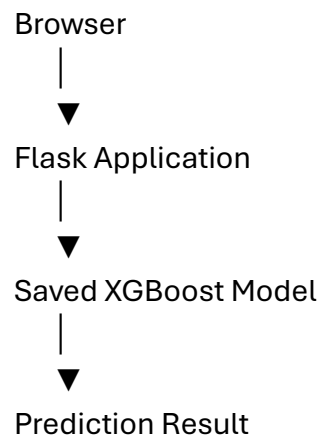


Machine Learning Workflow

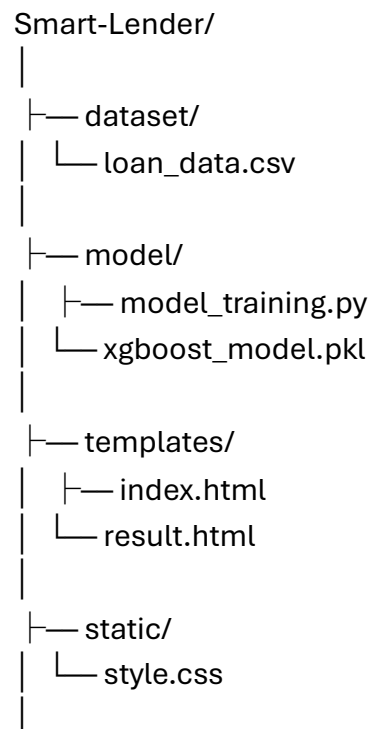




Deployment Structure



Directory Structure



```
├── tests/
│   └── test_prediction.py
├──
├── app.py
├── requirements.txt
└── README.md
```

Technologies Used

- Python
 - Flask
 - Pandas
 - NumPy
 - Scikit-learn
 - XGBoost
 - HTML
 - CSS
 - Bootstrap
 - IBM Cloud
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